Exercise 4 : SQL Practice exercise : Joins, union filtering and Aggregrate functions

1 SQL Joins : INNER JOIN

Retrieve all employees and thier assigned projects Using INNer Join Return: Employee_ID, first_name, Last_name, Department, Salary, Project_ID, Project_name Budget, Status.

Syntax

```
SELECT E.first_name, E.Last_name, E.Department, E.Employee_ID,
E.Salary, P.Project_ID, P.project_Name, P.Budget, P.Status
FROM Workspace.default.table_1_employees_Sql_Join_1 AS E
INNER JOIN Workspace.default.table_2_projects_Sql_Join_1 AS P
ON E.Employee_ID = P.Employee_ID;
```

	first_name	Last_name	Department	Employee_ID	Salary	Project_ID	Project_Name
1	John	Doe	IT	1	70000	101	AI Development
2	Alice	Smith	HR	2	60000	102	Employee training
3	John	Doe	IT	1	70000	103	CyberSecurity Audi
4	Bob	Johnson	finance	3	75000	104	finance Analysis
5	Emma	Wilson	Sales	5	65000	105	Market Expansion
6	Michael	Clark	finance	6	80000	106	Risk Management

table Continuation

	Budget	Status
1	100000	Completed
2	50000	ongoing
3	75 000	Pending
4	9 0000	ongoing
5	65 000	Completed
	8 0000	Pending

2. Left Join

Retrieve All employees and their assigned Projects, including Employees Who Have no projects Using Left Join

Expected Return : first_name, Last_name, Department, Salary, Project_ID, Project_name, Budget, Status.

Syntax

```
SELECT A.Employee_ID, A.First_name, A.Last_Name, A.Department,
A.Salary, B.Project_Id, B.Project_Name, B.Budget, B.Status
FROM Workspace.default.table_1_employees_Sql_Joins1 AS A
LEFT JOIN Workspace.default.table_2_projects_Sql_Joins_1 AS B
ON A.Employee_ID = B.Employee_ID;
```

	Employee_ID	First_name	Last_name	Department	Salary
1	1	John	Doe	IT	70000
2	2	Alice	Smith	HR	60000
3	3	Bob	Johnson	finance	75000
4	4	David	Brown	IT	72000
5	5	Emma	Wilson	Sales	65000
6	6	Michael	Clark	Finance	80000
7	1	John	Doe	IT	70000

Continuation of table.

	Project_id	Project_name	Budget	Status
1	103	CyberSecurity Audit	75000	Pending
2	102	Employee Training	50000	Ongoing
3	104	finance Analysis	90000	Ongoing
4	NULL	Null	Null	Null
5	105	Market Expansion	65000	Completed
6	106	Risk Management	80000	Pending
7	101	AI Development	100000	Completed

3. Right JOIN

Retrieve all projects and their assigned employees, including project that have no employees using a Right Join
Return : Project ID, ProjectName, Budget, Status, Employee ID, firstName, Last Name, Department, Salary.

Syntax

```
SELECT tbl2. Project_ID, tbl2.Project_Name, tbl2. Budget,
tbl2.Status, tbl1. Employee_ID, tbl1.first_Name, tbl1. Last_name
tbl1. Department, tbl1. Salary.
RIGHT JOIN Workspace.default.table_2_projects_Sql_Join_1 As
tbl2
ON tbl1. Employee_ID = tbl2. Employee_ID;
```

	Project-ID	Project_Name	Budget	Status
1	101	AI Development	100000	Completed
2	102	Employee Training	50000	Ongoing
3	103	CyberSecurity Audit	75000	Pending
4	104	finance Analysis	90000	Ongoing
5	105	Market Exponsion	65000	Completed
6	106	Risk Management	80000	Pending

	Employee-ID	First_name	Last_name	Department	Salary
1	1	John	Doe	IT	70000
2	2	Alice	Smith	HR	60000
3	1	John	Doe	IT	70000
4	3	Bob	Johnson	Finance	75000
5	5	Emma	Wilson	Sales	65000
6	6	Michael	Clark	Finance	80000

4 full Outer Join

Retrieve all employees and projects including those without a Match in either table using a full outer Join
Return: EmployeeID, firstname, Lastname, Department, Salary
Project_ID, ProjectName, Budget, Status.

Syntax

```
SELECT A.Employee_ID, A.first_Name, A.Department, A.Salary,
B.Project_ID, B.Project_name, B.Budget, B.Status
Full OUTER JOIN Workspace.default.table_1_employees_Sql_Join_1 As A
from Workspace.default.table_2_project_Sql_Join_1 As B
ON A.Employee_ID = B.Employee_ID;
```

	Employee_ID	first_Name	Last_name	Department	Salary
1	3	Bob	Johnson	Finance	75000
2	4	David	Brown	IT	72 000
3	6	Michael	Clark	finance	80000
4	2	Alice	Smith	HR	6 0000
5	5	Emma	Wilson	Sales	65000
6	1	John	Doe	IT	70000
7	1	John	Doe	IT	70000

	Project_ID	Project_Name	Budget	Status
1	104	finance Analysis	90000	Ongoing
2	Null	Null	Null	Null
3	106	Risk Management	8 0000	Pending
4	102	Employee Training	5 0000	Ongoing
5	105	Market Expansion	65 000	Completed
6	103	CyberSecurity Audit	75 000	Pending
7	101	AI Development	100000	Completed

2.5 UNION & UNION ALL

Retrieve a list of all unique Cities Where employees are located and project Statuses Return:
Location (Rename the Column to location using an alias

```
SELECT Status As Location
FROM Workspace.default.table_2_project_Sql_join_1
UNION
SELECT City As Location
FROM Workspace.default.table_1_employees_Sql_Join_1;
```

	Location
1	Completed
2	ongoing
3	Pending
4	New york
5	Los Angeles
6	Toronto
7	London
8	Sydney

2.6 Union ALL

Retrieve a list of all Cities where employees are Located and project Statuses, allowing duplicates
Return : Location (Rename the Column to location using an alias.

Syntax

```
SELECT Status As Location
fRom Workspace. default. table_2_project_Sql_Join_1
union ALL
SELECT City As location
fRom Workspace. default. table_1_employees_Sql_Join 1;
```

output

	Location
1	Completed
2	ongoing
3	Pending
4	New york
5	Los Angeles
6	Toronto
7	London
	Sydney

	Employee_ID	firstname	Last_name	Salary	Department
1	3	Bob	Johnson	75 000	finance
2	4	David	Brown	72 000	IT
3	6	Michael	Clark	80000	finance

Output

3.7 filtering Statement : Retrieve employees who earn More than 70 000.

Return : Employee ID, first_name Last_name Department, Salary (Syntax)

```
SELECT Employee_ID, first_Name, Last_Name, Department, Salary
fRom Workspace. default. table_1_employees_Sql_Join_1
Where Salary > 70000;
```

3.8 Retrieve employees Working in either IT or finance departments. Return : Employee ID, first Name, Last Name Department, Salary.

Syntax

```
SELECT Employee_ID, first_Name, Last_Name, Department, Salary
FROM Workspace.default.table_1_employees_Sql_JOIN_1
WHERE Department = 'IT' OR Department = 'finance';
```

	Employee_ID	First_name	Last_name	Department	Salary
1	1	John	Doe	IT	70000
2	3	Bob	Johnson	finance	75000
3	4	David	Brown	IT	72000
4	6	Michael	Clark	Finance	80000

3.9 Retrieve projects that are not yet Completed Return : Project ID, Project Name, Budget, Status.

Syntax

```
SELECT Project_ID, Project_Name, Budget, Status
FROM Workspace.default.table._2_project_Sql_JOIN_1
WHERE Status != 'Completed';
```

	Project_ID	Project_Name	Budget	Status
1	102	Employee Training	50000	Ongoing
2	103	CyberSecurity Audit	75 000	Pending
3	104	finance Analysis	90000	Ongoing
4	106	Risk Management.	80000	Pending

3.10 Retrieve Project that have a budget greater than 70 000 and are not Completed.

Return: Project ID, Project Name, Budget, Status.

Syntax

```
SELECT Project_ID, Project_Name, Budget, Status.
FROM Workspace.default.table_2_projects_Sql_Join_1
WHERE Budget > 70000 AND Status != 'Completed';
```

	Project_ID	Project_Name	Budget	Status
1	103	CyberSecurity Audit	75 000	Pending
2	104	finance Analysis	9 0000	Ongoing
3	106	Risk Management	8 0000	Pending

3.11 Retrieve employees from New york OR Toronto, Ordered by Salary in descending Order

Return: Employee ID, first Name, Last Name, Department, Salary City.

Syntax

```
SELECT Employee_ID, First_Name, Last_name, Department,
Salary, City
FROM Workspace.default.table_1_employees_Sql_JOIN_1
WHERE City IN ('New York', Toronto')
```

	Employee_ID	firstname	Last_Name	Department	Salary	City
1	1	John	Doe	IT	7 0000	New york
2	3	Bob	Johnson	finance	75 000	Toronto
3	6	Michael	Clark	finance	8 0000	New York

3.12 Retrieve the top 3 highest-paid employees.
Return: EmployeeID, First-Name, Department, Salary
Syntax.

```
SELECT Employee_ID, first_NAme, Last_Name, Department,
Salary
FROM Workspace.default.table_1_employees_Sql_Join_1
ORDER BY Salary DESC
Limit 3;
```

	~~First_Name~~	~~Last_Name~~	~~Department~~	~~Salary~~	~~City~~
~~1~~	~~John~~	~~Doe~~	~~IT~~	~~80000~~	
~~2~~					
~~3~~					

	Employee_ID	First_name	Last_name	Department	Salary
1	6	Michael	Clark	Finance	80000
2	3	Bob	Johnson	Finance	75 000
3	4	David	Brown	IT	72 000
4					

4.13 Aggregate functions With Group By AND Having
Find the total Salary per department Sorted in descending order Return: Department, total_Salary (Rename Sum (Salary Salary) as Total Salary
Syntax

```
Select Department,
Sum (Salary) As Total Salary
FROM Workspace.default.table_1_employees_Sql_Join_1
Group By Department
ORDER BY Total Salary DESC;
```

	Department	Total Salary
1	Finance	155000
2	IT	142 000
3	Sales	65 000
4	HR	60000

4.14 find the average Salary per City, but only include Cities Where Salary is greater than 65, 000

Return: City, Average Salary (Rename AVG (Salary) as Average Salary)

Syntax

```
Select City,
   AVG (Salary) As Average_Salary
FROM Workspace.default.table_1_employees_Sql_Join_1
Group By City
HAVING AVG (Salary) > 65 000;
```

	City	Average_Salary
1	New York	75 000
2	Toronto	75 000
3	London	72 000

4.15 Count the number of employees per department, including only department With More than 1 employee.

Return: Department, Employee Count (Rename Count (employeeID) as Employee Count).

Syntax

```
SELECT Department
  Count (Employee_ID) As Employee Count
from Workspace.default.table_1 employee_Sql_Join_1
Group By Department
HAVING Count (Employee_ID > 1;
```

	Department	Employee Count.
1	IT	2
2	finance	2

4.16 Retrieve the number of projects per Status but only includes Statuses With atleast 2 projects

Return : Status, Project Count (Rename Count (project ID project Count).

Syntax.

```
SELECT Status
Count (Project-ID) As Project Count
FROM Workspace.default.table-2-project-Sql-Join-1
Group By Status.
HAVING Count (Project-ID) >= 2;
```

	Status	Project Count
1	Completed	2
2	ongoing	2
3	Pending.	2

4.17 Retrieve the total project budget per employee, but only for employees Who are Managing Projects Worth More than 150, 000

Return : Employee ID, first-Name, Last-Name, Total Project Budget (Rename Sum (Budget) as Total Project Budget

Syntax

```
SELECT e. Employee-ID, e.first-Name, e. Last-Name,
Sum (p. Budget) As Total Project Budget
From Workspace. default. table-1-employees-Sql-join As e
JOIN Workspace. default. table-2-projects-Sql-Join-1 As p
ON e. Employee-ID = p. Employee-ID
Group By e. Employee-ID, e. first-name, e. Last-name
Having Sum (p. Budget) > 15 0000;
```

	Employee-ID	First-name	Last-name	Total Project budget
1	1	John	Doe	175000